

Scientists Link Gene to Human Speech

A new study suggests the beginnings of human speech are linked to genetics. The research identifies a protein – found only in people – that may have helped early humans develop spoken communication.

Scientists involved in the study say this new speaking ability became important for humans' survival. For example, speech permitted individuals to share information, organize activities and pass down knowledge. These abilities are now seen as an advantage humans had over their relatives, such as the Neanderthals and Denisovans.

The researchers recently published their findings in a study in the journal *Nature Communications*.

Liza Finestack teaches about speech and hearing at the University of Minnesota. She told *The Associated Press (AP)* the new study is "a good first step to start looking at the specific genes" that may affect speech and language development. Finestack was not involved in the study.

Dr. Robert Darnell has long been studying the protein – called NOVA1 – at his laboratory at New York's Rockefeller University. He helped lead the new research and was a writer of the study. Darnell told the AP the genetic version, or variant, included the protein that helped humans develop into the "dominant species" that remains today.

The latest research involved scientists using CRISPR gene editing methods to replace the NOVA1 protein found in mice with the one found in humans. The aim was to test the real-life effects of the genetic variant. The researchers were surprised to learn that the variant changed the way the animals called out to each other.

Baby mice with the human variant made a different sound than normal mice do when their mother came around. Adult male mice with the variant also made different sounds when they were near a female they wanted to mate with.

Darnell said both of these situations gave the mice a reason to speak. Those with the human variant "spoke differently," demonstrating the gene's influence in speech, he added.

This is not the first time a gene has been linked to speech. In 2001, British scientists said they had discovered the first gene tied to a language and speech disorder.

Researchers called this human language gene FOXP2. But even though FOXP2 was found to be linked to human language, the variant in modern humans was not found in our species alone. Later research found it was shared with Neanderthals. The NOVA1 variant in modern humans, on the other hand, is found only in our species, Darnell said.

The presence of a gene variant is not the only thing that permits people to speak. The ability also depends on physical elements in the human throat and areas of the brain that work together to help people produce speech and understand language.

Darnell said he hopes the recent work can lead to new ways to treat speech-related problems.

The University of Minnesota's Finestack noted the genetic findings could also one day permit scientists to identify people who might need help developing speech and language early in life.

"That's certainly a possibility," she said.

I'm Jill Robbins.

Laura Ungar reported this story for the Associated Press. Jill Robbins adapted it for Learning English.

Vocabulary

advantage - n. ข้อได้เปรียบ, ประโยชน์

specific - adj. เฉพาะเจาะจง, เป็นพิเศษ

variant - n. รูปแบบที่แตกต่าง, สายพันธุ์ย่อย

dominant - adj. เหนือกว่า, มีอิทธิพลเด่น, ครบงำ

species - n. ชนิด, สปีชีส์ (ในทางชีววิทยา)

CRISPR - n. คริสเปอร์ (เครื่องมือตัดต่อยีนที่ช่วยให้นักวิทยาศาสตร์ตัดและแก้ไขลำดับ DNA ได้อย่างแม่นยำ)

throat - n. ลำคอ, คอหอย

Reading Comprehension Quiz

1. What is the primary focus of the new study mentioned in the article?

The development of early human tools.

The link between human genetics and the beginnings of speech.

The differences in communication between humans and Neanderthals.

The discovery of new proteins in the human body.

2. According to the study, how did the ability to speak benefit early humans?

It allowed them to build more advanced shelters.

It improved their hunting techniques dramatically.

It facilitated sharing information, organizing activities, and passing down knowledge.

It helped them to become physically stronger than other species.

3. In which publication were the findings of the researchers recently published?

The Associated Press

Rockefeller University Journal

Nature Communications

University of Minnesota Science Review

4. What was Liza Finestack's role in relation to the study?

She was the lead researcher of the study.

She taught about the study at the University of Minnesota.

She provided an expert opinion on the study, despite not being involved in it.

She helped publish the findings in Nature Communications.

5. According to Dr. Robert Darnell, what significant role did the genetic variant including the NOVA1 protein play?

It allowed humans to develop stronger immune systems.

It helped humans evolve into the dominant species.

It primarily improved human problem-solving skills.

It enabled early humans to migrate across continents.

6. What experimental method did scientists use in the latest research to study the NOVA1 protein?

They observed human behavior over several generations.

They analyzed ancient human DNA samples.

They used CRISPR gene editing to replace mouse NOVA1 with human NOVA1.

They conducted extensive surveys on language development in children.

7. What was the initial surprising outcome when mice were engineered with the human NOVA1 variant?

They showed a significant increase in intelligence.

Their physical appearance changed noticeably.

They began to communicate in a different way.

They became more aggressive than normal mice.

8. How did baby mice with the human NOVA1 variant behave differently compared to normal mice?

They were more playful when their mother was around.

They made different sounds when their mother came near.

They learned to find food more efficiently.

They developed a stronger sense of smell.

9. What did Darnell conclude about the mice with the human variant that "spoke differently"?

The different sounds were coincidental and not significant.

The gene variant had influenced their ability to produce speech.

The mice were experiencing distress due to the gene editing.

They were attempting to mimic human language.

10. Before the NOVA1 study, what other significant discovery linked a gene to speech?

The identification of a gene responsible for human memory.

The discovery of a gene tied to a language and speech disorder in 2001.

Research linking a gene to the ability of birds to sing.

The finding of a gene that controls human hand movements for communication.

11. What is a key distinguishing characteristic of the NOVA1 variant compared to the FOXP2 gene?

FOXP2 is exclusive to modern humans, while NOVA1 is shared with Neanderthals.

NOVA1 is found only in modern humans, while FOXP2 was also found in Neanderthals.

Both NOVA1 and FOXP2 are found exclusively in modern humans.

Neither NOVA1 nor FOXP2 has been definitively linked to speech.

12. Besides a gene variant, what other elements are necessary for people to speak?

A specific diet rich in proteins.

Advanced social structures and writing systems.

Physical elements in the throat and specific brain areas working together.

Constant exposure to human speech from birth.

13. What is Dr. Darnell's hope regarding the future applications of the recent research?

To genetically engineer animals to speak human languages.

To discover new species of speaking animals.

To develop new treatments for speech-related problems.

To prove that all human behaviors are determined by genetics.

14. According to Liza Finestack, how could the genetic findings be used to help individuals in the future?

To identify and genetically modify those with speech disorders.

To predict a person's intelligence level based on their genes.

To identify people who might need early help with speech and language development.

To understand how different languages evolved over time.

15. Based on the statements from Darnell and Finestack, what is the overall outlook for the practical application of this research in the future?

The research is purely theoretical with no practical applications foreseen.

There is a strong possibility that it will lead to diagnostic tools and treatments for speech issues.

Its main purpose is to understand the history of human evolution, not current medical problems.

The findings are too controversial to be applied in real-world scenarios.